INCH POUND

MIL-C-3098/110B
5 May 1993

SUPERSEDING
MIL-C-3098/110A
2 November 1983

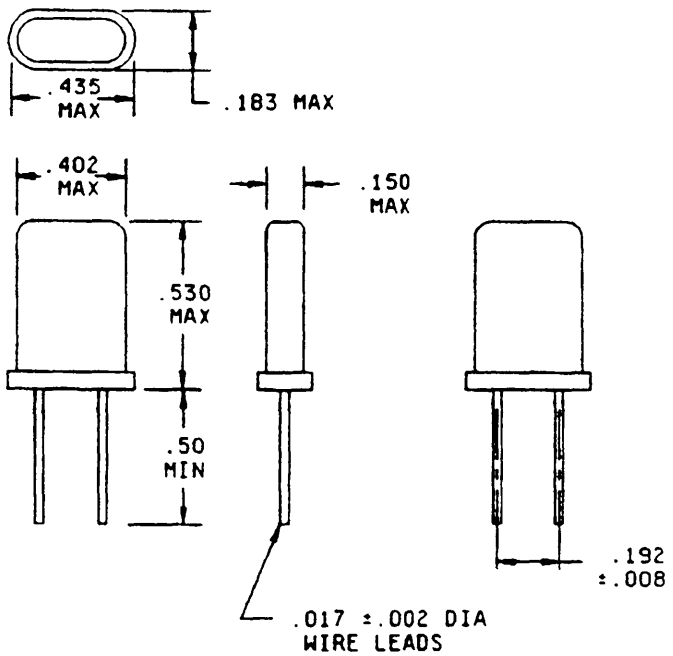
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR129/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 9 to 12 MHz; fundamental; noncontrolled; antiresonance.




| Inches | mm |
|---|---|
| .002 | .05 |
| .008 | .20 |
| .017 | .43 |
| .150 | 3.81 |
| .183 | 4.65 |
| .192 | 4.88 |
| .402 | 10.21 |
| .435 | 11.05 |
| .50 | 12.7 |
| .530 | 13.46 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR129/U.


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REQUIREMENTS:

Holder: HC-18/U (MIL-H-10056/9) or HC-43/U.

Dimensions, marking, and configuration: See figure 1.

Frequency range: 9.0 to 12.0 MHz, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Mode of oscillation: Fundamental.

Rated drive level: 5.0 ±1.0 mW.

Antiresonance, load capacitance: 32.0 ±0.5 pF.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±35 ppm.

Equivalent resistance:

| MHz | Ohms, maximum |
|---|---|
| 9.0 to 10.0 | 35 |
| 10.0+ to 12.0 | 25 |

Shock (specified pulse): Method 213 of MIL-STD-202; test condition K, consisting of 18 impact shocks of 15 g's each having a duration of 11 milliseconds; 3 shocks along each direction in each of 3 mutually perpendicular planes.

Frequency change permitted: ±5 ppm.

Equivalent resistance: ±15%, or 2 ohms, whichever is greater.

Vibration: Method 204 of MIL-STD-202; test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15%, or 2 ohms, whichever is greater.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15%, or 2 ohms, whichever is greater.

Aging:

Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.



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## CONCLUDING MATERIAL

Custodians:

Army - ER
Navy - EC
Air Force - 85

User activities:

Air Force - 17, 19, 84

Preparing activity:

Army - ER

Agent:

DLA - ES

(Project 5955-0678-07)



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